6.8 Cost of Programmability

Cost of Programmability

- The ***programmability*** in an FPGA comes w/ a considerable amount of ***hardware cost***.
  - In an **SRAM based FPGA**, ***SRAM*** is used for creating the ***logic blocks***, the ***programmable interconnects***, and the ***programmable I/O blocks***.

Logic Block

- Logic blocks in many modern FPGAs contains ***4-variable function generators*** and ***MUXs***:
  - 4-variable function generators (***LUT4s***) :
    - Each takes 16 bits of SRAM: logic function is realized by loading appropriate bits into the LUT
  - MUXs:
    - may be used to select b/t various ***generated functions***,
to choose b/t ***latched*** and ***unlatched outputs***,
to generate functions of ***more variables***.
    - One bit of SRAM is required to implement the ***select input*** of the 2-to-1 MUXs, and two bits of SRAM are required for select lines of the programmable 4-to-1 MUXs, ...

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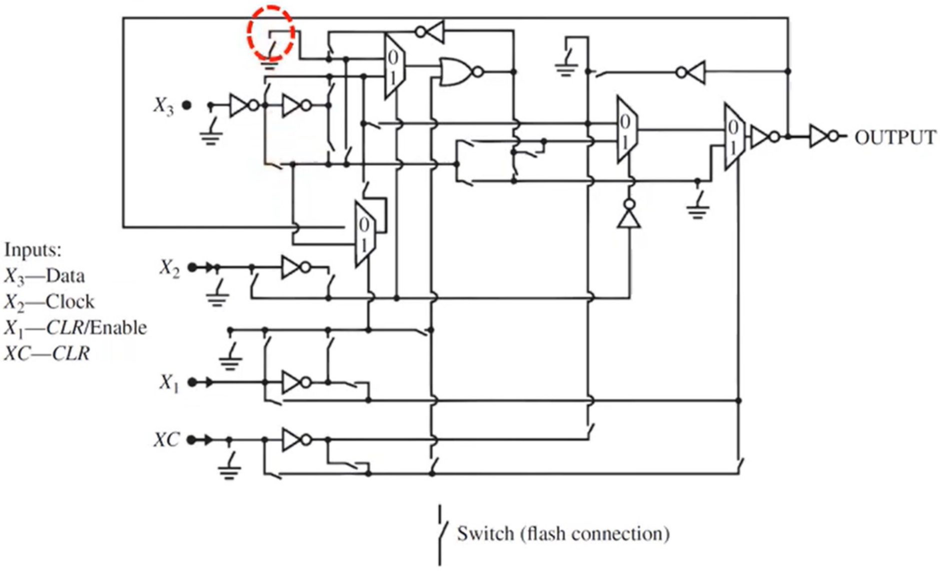
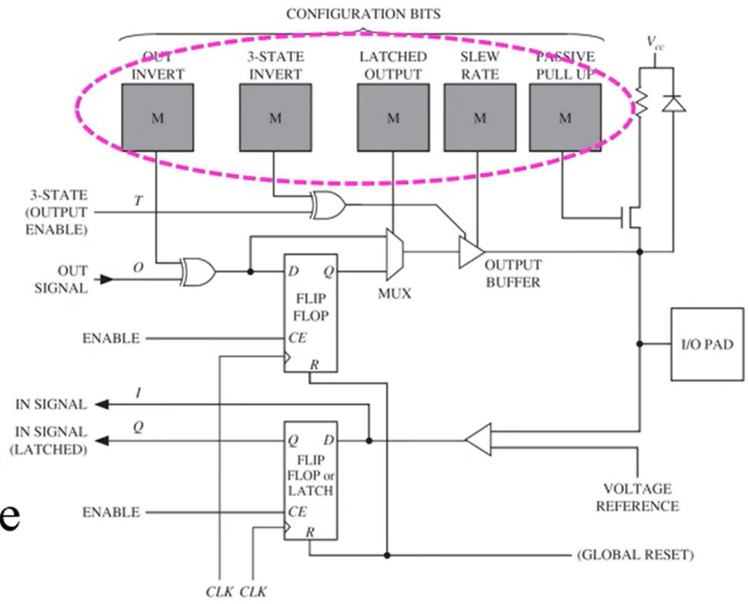
Example

- Microsemi Fusion FPGA:
  - each ***switch*** shown in the figure needs a ***flash memory cell***.



I/O Block

- I/O blocks:
  - contain several programmable points.
  - **Memory bits** are used to control the ***configuration***
    - to ***invert outputs***,
      to enable ***tristate output***,
      to enable the ***latching of output***,
      to control the ***slew rate*** of the signal,
      to enable ***pull-up resistors***, …



* M: memory cell

(Fig 3-39)

Cost of Programmability

- The ***flexibility*** and ***programmability*** of the FPGA comes at a high cost:
  - SRAM cell: typically takes 6 transistors
  - Flash memory cell: consumes ≈25% of an SRAM cell's area
- \# of ***configuration bits*** in example FPGAs:
  - P.371, Table 6-3

- # of ***configuration bits*** in example FPGAs:

Vendor	Device Family	Device	No. of Configuration Bits	No. of Logic Blocks	No. of LUTs	No. of Usable I/O Pins
Xilinx	Kintex 7	7K70T 7K40T	23 M 143 M	10,250 74,650	41,000 298,600	300 400
Xilinx	Artix 7	7A100T 7A200T	29 M 74 M	15,850 33,650	63,400 134,600	300 500
Xilinx	Virtex-6	XC6VLX75T XC6VLX760	26 M 177 M	11,640 118,560	46,560 474,240	360 120
Xilinx	Virtex-5	XC5VLX30 XC5VLX330	8.4 M 79.7 M	4,800 51,840	19,200 207,360	400 1200
Xilinx	Virtex-II	XC2V40 XC2V8000	0.3 M 26.2 M	256 46,592	512 93,184	88 1108
Xilinx	Spartan 3E	XC3S100E XC3S1600E	0.6 M 6.0 M	960 14,752	1,920 29,504	108 376
Xilinx	Spartan 6	XC6SLX4 XC6SLX150	2.7 M 33.9 M	600 23,038	2400 92,152	132 576
Altera	Stratix II	EP2S15 EP2S180	4.7 M 49.8 M	6,240 71,760	12,480 143,520	366 1170
Altera	Stratix	EP1S10 EP1S80	3.5 M 23.8 M	10,570 79,040	10,570 79,040	426 1238
Altera	Arria V	5AGXA1 5AGXB7	71 M 185.9 M	28,302 190,240	76,800 516,096	416 704
Altera	Arria II	EP2AGX45 EP2AGX260	29.6 M 86.9 M	18,050 102,600	45,125 256,500	364 612
Altera	Cyclone II	EP2C5 EP2C70	1.3 M 14.3 M	4,608 68,416	4,608 68,416	158 622